YAN ALMEIDA

Software Engineer · Backend, Systems & Industrial Software

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Summary

Chemical Engineer (**BSc from Budapest University of Technology and Economics — BME**) transitioning into **Software Engineering** via **ISPGAYA** (Porto, Portugal). Combines deep mathematical modeling, thermodynamics, and numerical analysis with enterprise backend software design in **C# (.NET 8)**, low-level systems in **C11**, and algorithmic telemetry in **Python**. Creator of an end-to-end industrial software ecosystem comprising 5 integrated repositories with 187 automated unit/integration tests and CI/CD pipelines.

Technical Skills

- **Languages:** C# 12 (.NET 8), C11 (GCC), Python 3.12 (Asyncio), SQL, Bash
- **Backend & Frameworks:** ASP.NET Core Web API, Entity Framework Core, Npgsql, PostgreSQL Binary `COPY`, System.Threading.Channels, SignalR WebSockets, FastAPI, Pydantic v2, RESTful APIs, Swagger/OpenAPI
- **Architecture & Systems:** Clean Architecture, SOLID Principles, C-ABI Foreign Function Interface (FFI via C# P/Invoke & Python `ctypes`), Directed Acyclic Graphs (DAGs), Kahn's Topological Sort, Numerical Solvers (Runge-Kutta 4, Newton-Raphson), Low-Level Memory Management
- **Data & Databases:** PostgreSQL 16 (Composite Indexing, Time-Series), SQLite (`aiosqlite`, ADO.NET), Relational Modeling
- **DevOps & Tooling:** Docker, Docker Compose, Git, GitHub Actions (Automated CI), Linux (Ubuntu), xUnit, Pytest

Featured Software Projects

IndustrialTelemetry.Engine — C# 12 · .NET 8 · PostgreSQL 16 · Clean Architecture · Docker

Enterprise Telemetry Ingestion, Batch State Machine, and Safety Alarm Engine

- Engineered high-throughput telemetry ingestion pipeline using bounded in-memory channels (`System.Threading.Channels`) with a hosted background worker processing 50,000 events/batch.
- Implemented direct zero-allocation PostgreSQL Binary `COPY` streaming (`NpgsqlBinaryImporter`) with composite indexing `(batch_id, timestamp_utc)`, drastically reducing ingestion latency.
- Implemented batch lifecycle state transitions (`Scheduled` → `InProgress` → `Completed/Aborted`) with compliance auditing and automated safety interlocks; validated with **41 xUnit automated tests**.

ChemFlow.NET — C# 12 · .NET 8 · Clean Architecture · SignalR · C11 Interop · SQLite

Modular Chemical Process & Rigorous Thermodynamic Simulation Engine

- Built thermodynamic engine featuring Peng-Robinson Cubic EOS, analytical Cardano solver, and safeguarded Rachford-Rice flash solver.
- Developed rigorous multicomponent staged Distillation Column (RadFrac MESH solver) using Thomas Algorithm (TDMA) and Holland's θ-Method, enforcing mass closure with error $\Delta m/m < 10^{-14}$.
- Integrated native C11 ODE reactor solver via P/Invoke and streamed real-time dynamic simulation progress over WebSockets via **ASP.NET Core SignalR**; covered by **59 xUnit tests**.

ProcessTwin.Py — Python 3.12 · FastAPI · Asyncio · C11 FFI · WebSockets · SCADA

Real-Time Industrial Process Digital Twin & SCADA Telemetry Engine

- Achieved a **9.4x speedup** (2.66M integration steps/s) by bridging to compiled C11 ODE library (`libreactorcore.so`) via `ctypes` FFI with zero bit-level IEEE-754 discrepancy against analytical benchmarks.
- Streamed 5 Hz live telemetry over WebSockets, emulated sensor noise on 4–20 mA transmitters, and evaluated Safety Instrumented System (SIS / IEC 61511) interlocks; verified with **48 pytest tests**.

ReactorCore.C — C11 · GCC · Makefile · Zero-Heap Numerical Engine

High-Performance Dynamic Reactor ODE Solver & Exported C-ABI Engine

- Authored classical 4th-Order Runge-Kutta (RK4) integrator for non-linear Arrhenius kinetics with zero heap allocations in numerical loops, exported via standard C-ABI for multi-language consumption (**27 C11 tests**).

Education

- **CTeSP in Information Systems Technologies and Programming (TPSI)** — 2024 – Present
  Instituto Superior Politécnico Gaya (ISPGAYA) — Vila Nova de Gaia / Porto, Portugal
- **BSc in Chemical Engineering** (Taught in English) — Budapest, Hungary
  Budapest University of Technology and Economics (BME) — Full Scholarship Awardee
- **Technical Diploma in Chemistry** — Rio de Janeiro, Brazil
  Instituto Federal do Rio de Janeiro (IFRJ)

Languages

Portuguese: Native | **English:** Full Professional Proficiency / C1 Advanced (BSc conducted in English)